

1 **Feasibility of Field-Derived High-Resolution Event Replay for Traffic-**
2 **Signal Controller Testing**

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7 Total manuscript pages: 10

Structured Abstract

2 **Objectives.** Manual controller testing cannot reproduce a full day of simultaneous detector, pedestrian, coordination,
3 and preemption activity across many production timing configurations. This study evaluates whether high-resolution
4 events recorded at live field controllers can be replayed to test controllers and automatically compared to identify
5 operational differences.

6 **Methods.** Approximately 23 hours of events from each of 25 production configurations were converted to National
7 Transportation Communications for Intelligent Transportation Systems Protocol calls and replayed to controller
8 emulators. Output events were grouped by timestamp and aligned with dynamic time warping using Jaccard distance
9 between discrete event sets. The software also checked clearance behavior, pedestrian and preemption states, and
10 863 defined incompatible movement pairs. Feasibility was evaluated with a software-release comparison and a same-
11 software timing-parameter intervention.

12 **Findings.** All 11 configurations unchanged between two version 2.18.1 campaigns passed, with 99.6 percent mean
13 sequence match and 97.7 percent mean timing match. Deliberate trailing-overlap changes were flagged in 11 of
14 14 affected configurations and no unchanged configurations. Version 2.15.1 versus 2.18.1 produced 19 aggregate
15 passes and six review cases. Detailed diagnostics confirmed correction of a rail-preemption bug, faster coordination
16 transition behavior, consistent handling of an expected dynamic red-clear extension, and no virtual conflicts in the
17 three campaigns.

18 **Novelty.** The study combines long-duration field-derived input replay with automated alignment and operational
19 screening of complete controller-output traces across numerous production configurations.

20 **Practical Applications.** Once configured, the open-source workflow can replace much manual input toggling with
21 repeatable release, timing-change, and incident-replication tests.

1 INTRODUCTION

2 Controller acceptance testing is commonly limited by staff time. Manually toggling detector, pedestrian, and preemption inputs can confirm individual functions, but it cannot practically reproduce a full day of interacting calls across 25 production timing configurations. Rare sequences, coordination transitions, and preemption recovery may therefore receive little coverage.

6 This project uses field operation itself as the test script. Approximately 23 hours of high-resolution events from each production controller are converted to National Transportation Communications for Intelligent Transportation Systems Protocol (NTCIP) calls and replayed to test controllers. The resulting controller events are compared with an accepted baseline. The central question is:

10 Can high-resolution field events be replayed to test controllers, and can the resulting output sequences be aligned and compared automatically to identify operational differences?

12 A useful system must do more than replay inputs. Unchanged controllers must produce outputs repeatable enough for automatic alignment, while missing, additional, or shifted events must remain visible. The system must also direct staff to operationally important findings—clearance irregularities, pedestrian or preemption differences, and incompatible outputs—without requiring manual inspection of millions of event records.

16 The evaluation covers 25 production configurations and three replay campaigns: a version 2.15.1 baseline, version 2.18.1, and version 2.18.1 after trailing-overlap timing changes. The contribution is a practical, open-source method for comprehensive behavioral regression testing. NTCIP, controller emulation, automated testing, high-resolution logging, and dynamic time warping (DTW) are established technologies; their use together with full-day field-derived traces is the focus of this study.

21 RELATED WORK

22 Li et al. and the Idaho Transportation Department developed automated controller testing that used XML scripts to activate inputs and verify predefined responses (Li et al. 2008; Ahmed et al. 2010). Tung extended NTCIP-based automated testing across controller devices (Tung 2012; 2015). These systems established the value of repeatable automated tests. The present method differs by deriving long-duration tests from field logs and using a prior complete output trace as the expected response rather than requiring staff to author every input and assertion.

27 Controller-in-the-loop studies have connected controllers and emulators to traffic simulation through physical or virtual interfaces (Wang et al. 2021). Stevanovic, Klanac, and Radivojevic evaluated high-resolution logging across six controller vendors (Stevanovic et al. 2017). Those studies support the use of controlled controller environments and show that output-event semantics require platform-specific validation. They did not evaluate field-log replay for software regression across production configurations.

32 High-resolution event records provide tenth-second controller and detector histories (Sturdevant et al. 2012). DTW provides a monotone alignment for sequences with local timing variation (Sakoe and Chiba 1978). The literature reviewed for this study did not identify a prior evaluation combining full-day field-derived input replay, automatic complete-output alignment, and repeated software and timing tests across numerous production configurations.

36 REPLAY AND TEST ARCHITECTURE

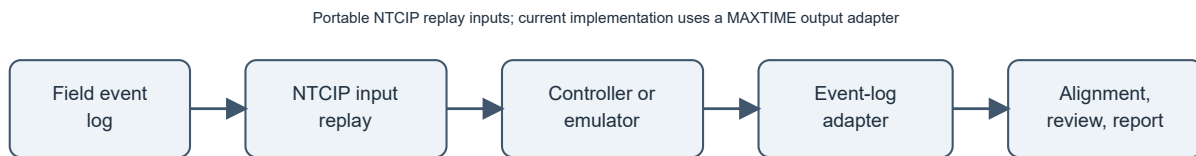


Figure 1: Field-derived testing workflow. Standard NTCIP objects form the replay-input boundary; controller output collection and event normalization require a product adapter.

1 Field event conversion

2 The 25 source logs span approximately 9:00 a.m. to 8:00 a.m. the next day and contain 7,296,218 records, or 574.998
3 configuration-hours. Replay selects vehicle detector, pedestrian detector, and preemption changes. Table 1 shows the
4 conversion implemented in the package.

Input	Event codes	Parameter	NTCIP replay state
Vehicle detector	81 off; 82 on	Detector 1–64	Update one bit in an eight-detector group and write the cumulative integer.
Pedestrian detector	89 off; 90 on	Detector 1–64	Update one bit in an eight-detector group and write the cumulative integer.
Preemption	104 off; 102 on	Preempt number	Write the corresponding Boolean preempt state.

5 Table 1. Conversion from field events to replay inputs.

6 Repeated on/off records are repaired by imputing the missing opposite transition. Dummy detector numbers 65 and
7 above are excluded. Before replay, all calls are reset to zero. The package then sends SNMP SET operations to NTCIP
8 1202 vehicle, pedestrian, and preempt objects at the recorded time of day (American Association of State Highway and
9 Transportation Officials and Institute of Transportation Engineers and National Electrical Manufacturers Association
10 2023). Preserving time of day also exercises coordination and time-of-day plan changes.

11 The evaluated implementation used MAXTIME emulators loaded with production databases. Emulator accommoda-
12 tions removed detector delay/extension, disabled unused input/output modules, removed inverted preempt logic, used
13 localhost peer addresses, and mapped overlap pedestrian calls to dummy detector inputs. Output events were collected
14 through a MAXTIME HTTP interface. A different controller family can use the same comparison method if it supports
15 the required NTCIP inputs and an adapter maps its output events to the common timestamp/event/parameter format.

16 Parallel replay and latency control

17 Signals assigned to available emulator targets run concurrently. The 25 configurations were processed in batches
18 over approximately one week without requiring staff to operate each input. After the initial setup, future releases are
19 expected to require only a few hours of staff preparation and review; this is operational experience, not a formal
20 labor study.

21 Replay timing includes a configurable latency offset. During the version 2.18.1 campaigns, the runner also estimated
22 latency from isolated detector-on events and updated the offset during collection. The stored release database contains
23 707,458 latency samples and 6,577 applied updates. The three campaign databases each contain 627,240 replay
24 commands and between 7.57 and 8.02 million controller-output events.

25 OUTPUT ALIGNMENT AND AUTOMATED CHECKS

26 Discrete-event dynamic time warping

27 Events at the same timestamp, or within 0.25 seconds, are grouped into sets of (event identifier, parameter)
28 pairs. Let A_i and B_j be reference and candidate event sets. Their local cost is Jaccard distance:

$$d(A_i, B_j) = 1 - \frac{|A_i \cap B_j|}{|A_i \cup B_j|}.$$

29 Identical event groups cost zero. Missing or additional events increase the cost. The cumulative DTW cost is

$$D(i, j) = d(A_i, B_j) + \min(D(i-1, j), D(i, j-1), D(i-1, j-1)),$$

30 with $D(0, 0) = 0$ and inaccessible borders set to infinity. Backtracking produces an order-preserving path; horizontal
31 and vertical path segments identify additions, omissions, or local shifts rather than hiding them.

Illustrative event grouping before sequence and timing comparison

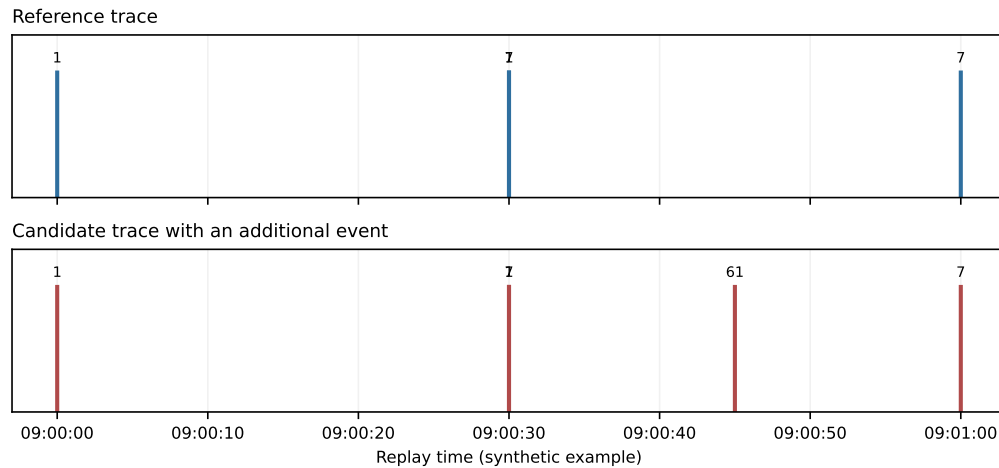


Figure 2: Simplified discrete-event alignment. DTW absorbs small timing shifts while missing and additional event groups retain nonzero costs.

1 Sequence scores are calculated in 45-minute windows advanced every 40 minutes. Sixty seconds are clipped from
2 each window edge to avoid penalizing a cycle split at a boundary. Sequence match is the percentage of aligned event-
3 group pairs with zero Jaccard distance. Timing match is the percentage of exactly matched groups within 0.50 seconds
4 after temporal alignment. A configuration passes at 95 percent sequence match and 90 percent timing match.

5 Vehicle phase-call similarity provides a separate replay-reliability check. Windows below 85 percent are displayed but
6 excluded from the configuration average. This helps distinguish controller differences from a test environment that
7 did not reproduce the intended inputs.

8 **Operational checks and virtual conflict monitor**

9 The automated review is broader than sequence scoring. It compares phase and overlap intervals, clearance durations,
10 pedestrian service, preemption, and coordination-transition states. It highlights short or irregular yellow/red clearances
11 and material changes in occurrence or duration.

12 A software-based virtual conflict monitor reconstructs active phase, overlap, pedestrian, and overlap-pedestrian states
13 from every collected output event. Each state is checked against configuration-specific incompatible pairs. The study
14 definitions contain 863 incompatible pairs across 21 configurations. No conflicts were stored in the 2.15.1, 2.18.1, or
15 parameter-modified 2.18.1 campaigns. This automated screen is a major increase in test coverage, but it complements
16 rather than replaces a cabinet conflict monitor or formal safety certification.

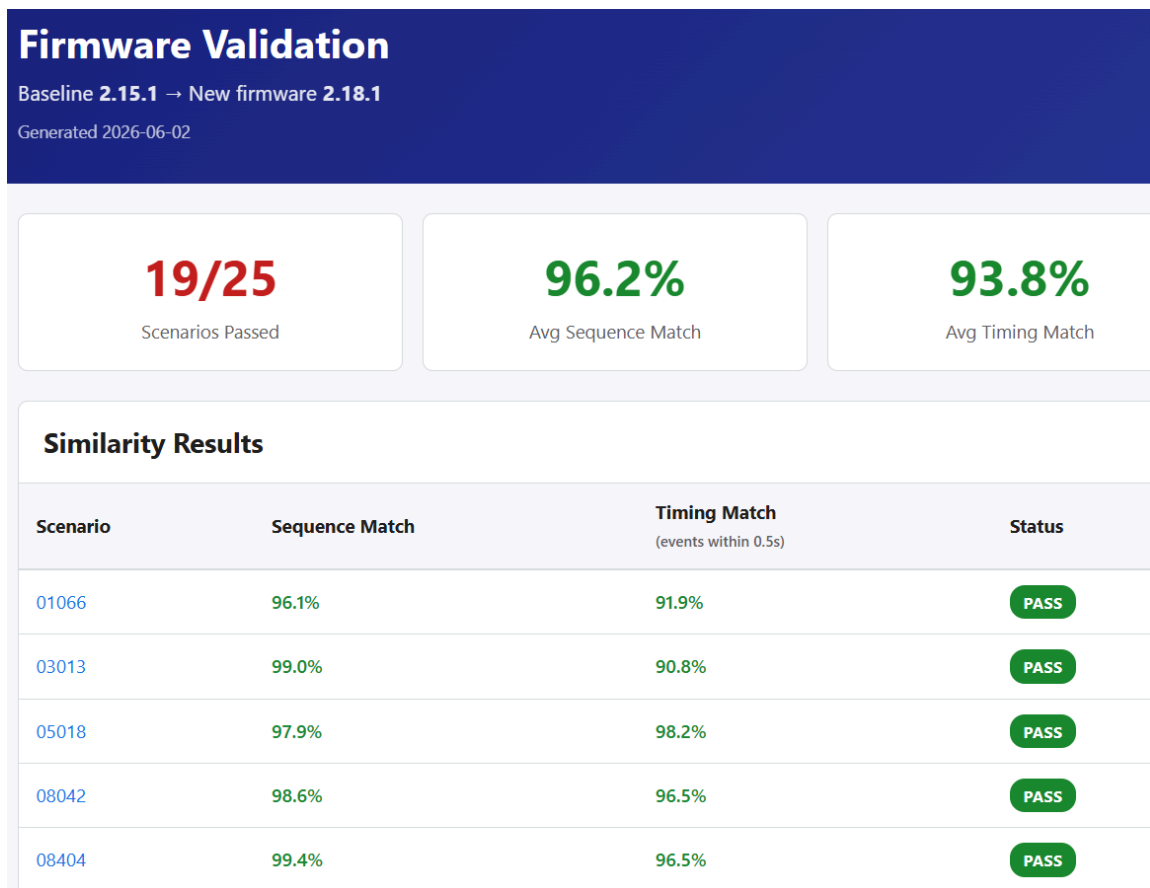


Figure 3: Example automated report. Aggregate sequence and timing results lead to focused views of clearance, pedestrian, preemption, transition, and conflict findings, allowing staff to review exceptions instead of the full event stream.

1 EXPERIMENTAL DESIGN

2 Table 2 separates the two tests. The release comparison represents normal acceptance testing. The same-software
3 comparison provides a repeatability group and a known operational intervention.

Test	Reference	Candidate	Configurations	Question
Software release	2.15.1	2.18.1	25	Can full-day outputs be aligned and meaningful re-release differences surfaced?
Timing intervention	2.18.1	2.18.1 with applicable trailing-overlap changes	14 changed; 11 unchanged	Are unchanged replays repeatable, and are deliberate changes detected?

4 Table 2. Evaluation design.

5 Transaction histories identify trailing-yellow, trailing-red, or associated red-revert edits in 14 configurations. Eleven
6 configurations had no relevant edit. A changed parameter is detectable only if the recorded day exercises the affected
7 overlap operation.

8 RESULTS

9 Replay was repeatable and responsive to change

10 All 11 unchanged configurations passed across the separate 2.18.1 campaigns. Their mean sequence match was 99.6
11 percent (range 99.1–100.0 percent), and mean timing match was 97.7 percent (range 90.6–99.7 percent). These are
12 separate full-day runs, not a file compared with itself. The result demonstrates that controller output was deterministic
13 enough for automated alignment.

1 Eleven of 14 configurations with applicable trailing-overlap changes were flagged, while none of the 11 unchanged
2 configurations was flagged. The three changed configurations that passed had sequence matches of 99.3, 99.9, and
3 100.0 percent; the recorded input apparently did not materially exercise the affected behavior. Table 3 summarizes the
4 result without implying exhaustive coverage of every possible timing change.

Group	Flagged	Passed	Total
Applicable trailing-overlap edit	11	3	14
No relevant edit	0	11	11

5 Table 3. Same-software repeatability and parameter-intervention results.

6 The intervention removed short and variable overlap-yellow intervals while otherwise retaining similar operation in
7 the reviewed examples (Figure 4). This shows how the same field trace can evaluate a timing correction as well as a
8 software release.

Ovlp 10 Yellow shown below is 1.00s below the median in 2.18.1 (2.15.1: 3.00s vs median 4.00s; 2.18.1: 3.00s vs median 4.00s)

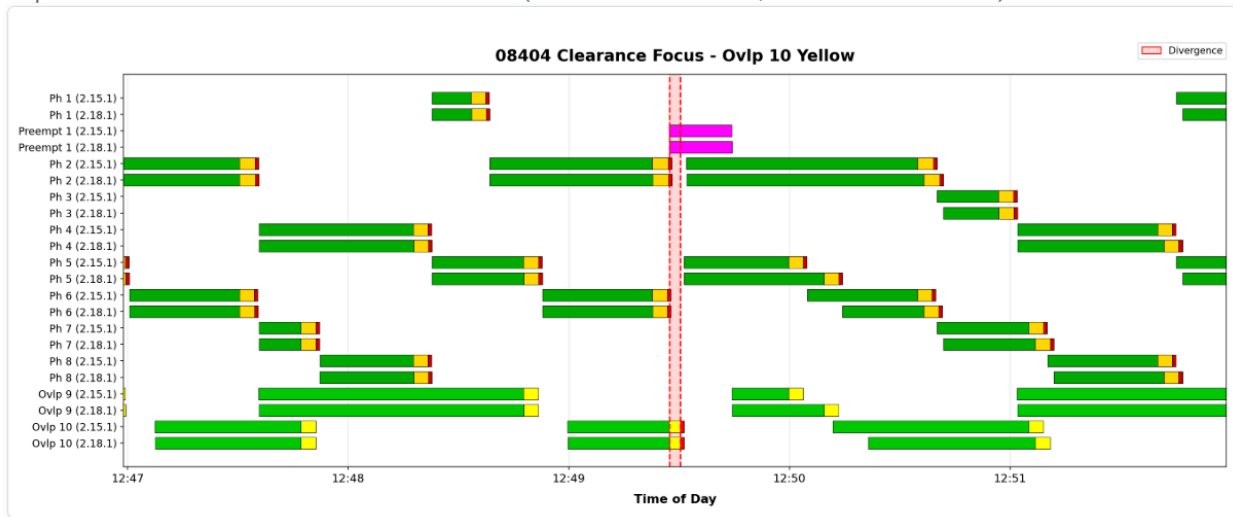


Figure 4: Representative parameter test. Adding trailing-yellow and trailing-red settings removed short, variable overlap-yellow intervals while the comparison checked the remainder of the replay for unintended changes.

9 Version 2.18.1 acceptance test

10 The release comparison produced 19 aggregate passes and six configurations requiring review. Mean sequence match
11 was 96.2 percent and mean reported timing match was 93.8 percent. Four configurations fell below the sequence
12 threshold; two exceeded the sequence threshold but fell below the timing threshold.

Outcome	Count	Configurations / interpretation
Passed both thresholds	19	Operationally similar overall; detailed diagnostics still identified localized changes.
Sequence review	4	12036, 2B045, 2B094, 2B339. The 12036 result likely reflects peer-to-peer operation not being configured correctly in the test environment.
Timing review	2	2B054 and 2B085.

13 Table 4. Version 2.15.1 versus 2.18.1 release-screen outcomes.

14 The detailed findings were more informative than the aggregate status:

- 15 • **Rail-preemption fix (13008).** Version 2.15.1 could leave Preempt 6 active with the agency's rail-preemption
16 configuration. Version 2.18.1 exited correctly (Figure 5). This confirmed a known bug fix.
- 17 • **Transition improvement (2B049).** Version 2.18.1 completed a short-way coordination transition faster, as shown
18 against the programmed split. The trace also raised a separate question: the controller appeared to be in step while
19 still reporting transition (Figure 6).

1 • **Expected clearance behavior (13008).** The report flagged an irregular phase 4 red-clearance interval. Review
2 showed that dynamic red-clear extension was configured, and both versions handled it consistently. The flag
3 demonstrated that an unexpected clearance change would have been visible.

4 The first two configurations passed the aggregate thresholds. This is important: the tool did not merely label controllers
5 pass or fail. It retained localized differences that confirmed a bug fix and exposed an algorithm change within otherwise
6 similar full-day operation.

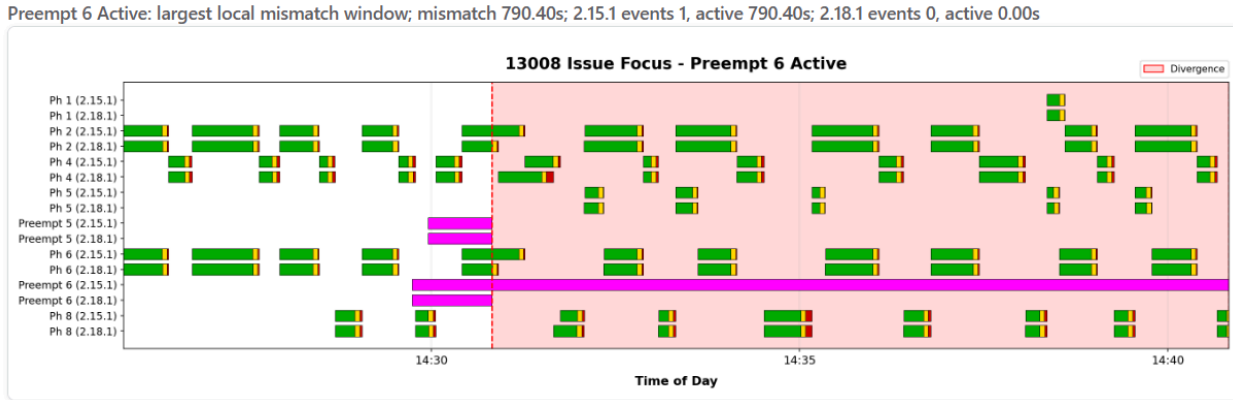


Figure 5: Configuration 13008, Preempt 6. Version 2.15.1 remained active under the rail-preemption configuration; version 2.18.1 exited correctly, confirming the known fix.

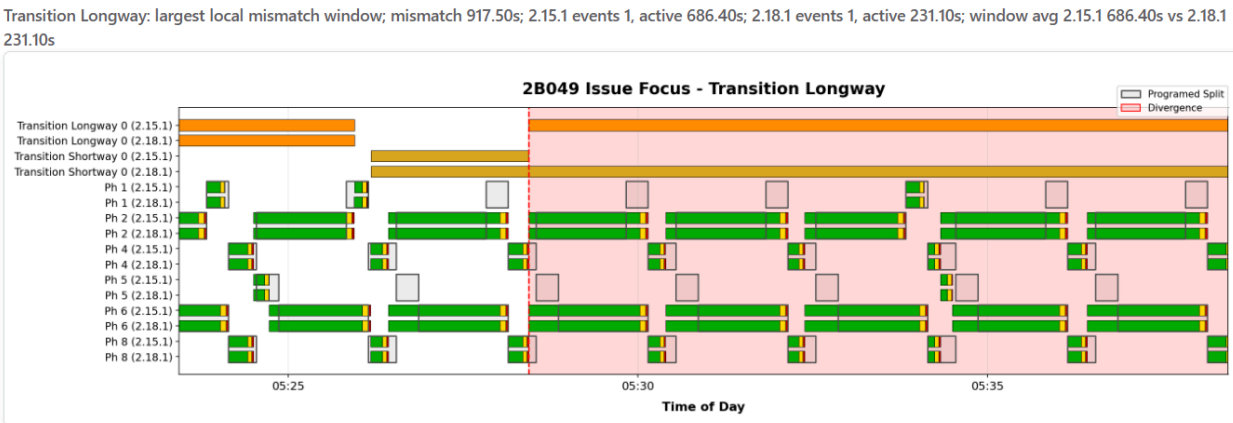


Figure 6: Configuration 2B049, short-way transition. Version 2.18.1 returned to the programmed split faster. The aligned trace also showed the controller reporting transition after appearing to be in step.

7 No new operation-impacting software defect was confirmed in version 2.18.1 from the reviewed results. The new
8 version was otherwise operationally similar across the tested configurations, and the virtual conflict monitor recorded
9 no incompatible simultaneous outputs.

10 DISCUSSION

11 The practical advantage is coverage. Staff do not need to toggle one input at a time and decide which combinations to
12 try. The system replays hundreds of thousands of calls across production timing configurations, runs available emula-
13 tors concurrently, and automatically evaluates millions of output events. Review effort is concentrated on exceptions.

14 Field-derived replay also preserves combinations that are difficult to invent in advance: simultaneous detector and
15 pedestrian calls, time-of-day plan changes, coordination transitions, and preemption during prevailing traffic operation.
16 A rare failure or flash event can be saved as a targeted trace, replayed to reproduce the problem, and used to test a
17 vendor correction or timing workaround. Scripted requirement tests remain appropriate for conditions that have not
18 occurred in the recorded data; field replay supplies breadth, not exhaustive state-space coverage.

1 A stored baseline acts as a behavioral oracle, but it is not assumed correct forever. The 13008 example correctly
2 appeared as a difference because the candidate fixed a baseline defect. Engineering review must classify each difference
3 as beneficial, harmful, expected, environmental, or unexplained before a new baseline is accepted.

4 The package is open source and separates standard inputs from product-specific outputs. Its NTCIP input interface can
5 be adapted to any controller that implements the required objects. A new controller family still needs an output collector,
6 event-code mapping, and validation of timestamp behavior. The current MAXTIME implementation demonstrates the
7 method; it is not a claim of adapter-free compatibility.

8 **LIMITATIONS**

9 The campaigns used one agency's production configurations and one controller-family emulator. Emulator accommo-
10 dations and incomplete peer-to-peer setup affected generalizability and likely explain the 12036 mismatch. Hardware
11 and a second controller family should be evaluated in future work.

12 A 23-hour trace tests only behavior exercised during that day. Three parameter-modified configurations produced
13 essentially unchanged output, illustrating the need to retain multiple field days and targeted incident traces. The
14 thresholds are engineering screening values rather than statistically optimized decision boundaries, and the planned
15 one-factor sensitivity analysis has not yet been run.

16 Five release-threshold review cases remain incompletely classified after accounting for the likely 12036 environment
17 issue. That does not invalidate the feasibility result, but final operational disposition would strengthen the acceptance-
18 test case study. The virtual conflict monitor is also limited to the incompatible pairs defined for 21 configurations and
19 is not a substitute for certified cabinet monitoring.

20 The controller replay requires emulator software, configuration databases, and agency infrastructure. The open-source
21 repository and proposed event bundle can reproduce the comparison and reporting stage without that equipment, but
22 not the original controller execution.

23 **CONCLUSIONS**

24 Field-derived high-resolution replay is feasible for comprehensive controller regression testing. Separate unchanged
25 version 2.18.1 runs averaged 99.6 percent sequence match and 97.7 percent timing match, showing that DTW could
26 reliably align the outputs. The method flagged 11 of 14 configurations with applicable timing changes and none of
27 the unchanged configurations. The version acceptance test automatically surfaced a rail-preemption bug fix, a coordi-
28 nation-transition improvement, an expected clearance irregularity, threshold review cases, and zero virtual conflicts
29 across the configured incompatible pairs.

30 The method replaces much manual input toggling with a reusable operational test corpus. Once the environment is
31 established, staff can test many configurations in parallel, allow the automated replay to run unattended, and review a
32 focused set of differences. The same framework supports software releases, timing changes, and reproduction of field
33 failures. Its open-source NTCIP design provides a practical path for other agencies and controller families to adapt the
34 approach.

35 **ACKNOWLEDGMENTS**

36 The author thanks Chris Primm, State Traffic Operations Engineer, Oregon Department of Transportation, for
37 reviewing the results and contributing ideas for testing.

38 **CODE AND DATA AVAILABILITY**

39 Signal-Replay is an open-source Python package available at [https://github.com/ShawnStrasser/ATC-Signal-](https://github.com/ShawnStrasser/ATC-Signal-Replay)
40 [Replay](https://github.com/ShawnStrasser/ATC-Signal-Replay). It includes replay and comparison code, tests, documentation, and an offline example. The final submission
41 should cite release v0.2.0 or the exact public commit. A comparison-only data bundle can include replay inputs,
42 collected outputs, parameters, and expected results without controller firmware or proprietary configuration databases.

43 **GENERATIVE-AI DISCLOSURE**

44 OpenAI Codex was used to inspect repository materials, assist with literature discovery, revise code and documen-
45 tation, generate visualizations from stored results, and draft and format the manuscript. The author reviewed the source
46 materials and is responsible for the methods, accuracy, interpretation, citations, and submitted text.

47 **REFERENCES**

48 Ahmed, Sk. Monsur, Cody Browne, Ahmed Abdel-Rahim, and Richard Wall. 2010. *An Automated Testing Tool for*
49 *Traffic Signal Controller Functionalities*. Technical Report FHWA-ID-10-180. [https://rosap.ntl.bts.gov/view/dot/](https://rosap.ntl.bts.gov/view/dot/23916)
50 23916.

- 1 American Association of State Highway and Transportation Officials and Institute of Transportation Engineers and
2 National Electrical Manufacturers Association. 2023. “NTCIP 1202 V03b: Object Definitions for Actuated Traffic
3 Signal Controllers Interface.”. <https://www.ntcip.org/document-numbers-and-status/>.
- 4 Li, Zhen, Ahmed Ahmed-Abdel-Rahim, Brian Johnson, and Michael Kyte. 2008. “Design of Traffic Controller
5 Automated Testing Tool.” *Transportation Research Part C: Emerging Technologies* 16 (3): 277–93. <https://doi.org/10.1016/j.trc.2007.10.001>.
- 6
7 Sakoe, Hiroaki, and Seibi Chiba. 1978. “Dynamic Programming Algorithm Optimization for Spoken Word Recog-
8 nition.” *IEEE Transactions on Acoustics, Speech, And Signal Processing* 26 (1): 43–49. [https://doi.org/10.1109/](https://doi.org/10.1109/TASSP.1978.1163055)
9 TASSP.1978.1163055.
- 10 Stevanovic, Aleksandar, Ivica Klanac, and Danilo Radivojevic. 2017. *Development of Minimum Standards for Event-*
11 *Based Data Collection Loggers and Performance Measure Definitions for Signalized Intersections*. Technical
12 Report Nos. BDV27–977–6. <https://rosap.ntl.bts.gov/view/dot/31807>.
- 13 Sturdevant, James R., Tim Overman, Eric Raamot, et al. 2012. *Indiana Traffic Signal Hi Resolution Data Logger*
14 *Enumerations*. Technical report. <https://doi.org/10.4231/K4RN35SH>.
- 15 Tung, Leonard. 2012. *Development of Automated Testing Tools for Traffic Control Signals and Devices (NTCIP and*
16 *Security)*. Technical report. <https://rosap.ntl.bts.gov/view/dot/25053>.
- 17 Tung, Leonard. 2015. *Development of Automated Testing Tools for Traffic Control Signals and Devices (NTCIP and*
18 *Security), Phase 2*. Technical Report Nos. BDV30–977–5. <https://rosap.ntl.bts.gov/view/dot/28634>.
- 19 Wang, Daobin, Zongbei Tian, and Guangchuan Yang. 2021. “Virtual Controller Interface Device for Hardware-in-the-
20 Loop Simulation of Traffic Signals.” *IEEE Intelligent Transportation Systems Magazine* 13 (2): 201–16. [https://](https://doi.org/10.1109/MITS.2019.2898968)
21 doi.org/10.1109/MITS.2019.2898968.